

Monthly Intelligence Report

Edition 001 — The Waikato River Valley Corridor

Where the energy transition is outrunning the grid. Inaugural edition · April 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

1,558

200 HV · 1,358 MV · 1,100 MV/LV

GRID LINES MAPPED

~12,400

~42,000 km coverage

MC SIMULATIONS

78.9 M

10,000 per substation

PUBLIC DATA SOURCES

28+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 19,800+ source data points per run, executes 3.1 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 40.5 million arithmetic operations — producing 63,576 output data points with full uncertainty quantification across 1,558 substations and ~1,686 grid lines spanning ~12,000 km.

Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering New Zealand's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Auckland_1341096064

Auckland, Auckland · 11 kV

0.566

High

0.141

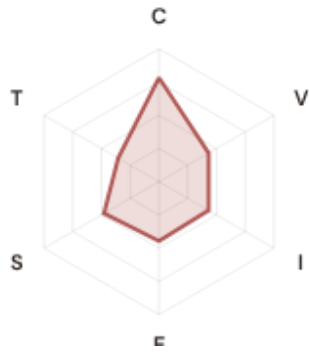
89th

R_FINAL

CLASSIFICATION

CI WIDTH

FLEET PERCENTILE



COMPONENTS

C Continuity: 0.782 / 0.30 (261%) ▲

I Infrastructure: 0.432 / 0.25 (173%)

S Saturation: 0.478 / 0.20 (239%)

V Voltage: 0.431 / 0.10 (431%)

E Economic: 0.447 / 0.10 (447%)

T Transition: 0.355 / 0.05 (709%)

MODIFIERS

R3 Consequence: 1.020 ↑ amplifies

R4 Graph Criticality: 0.990 → neutral

R6a Restoration: 1.000 → neutral

R6b Seismic: 1.033 ↑ amplifies

R7 Digital Readiness: 0.997 → neutral

This month's spotlight — automatically selected for April 2026 — is **Auckland_1341096064** in Auckland, Auckland. With an R_final of 0.566 (High band, 89th fleet percentile), its risk profile is dominated by the T (Transition) component at 709% of its weight ceiling, followed by E (Economic) at 447%. Modifiers collectively shift R_base by 4.0%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that New Zealand Numérique / New Zealand 2030 digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

184

High/Critical + R7 < median

AVG R7 MODIFIER

0.9965

Range [0.99, 1.05]

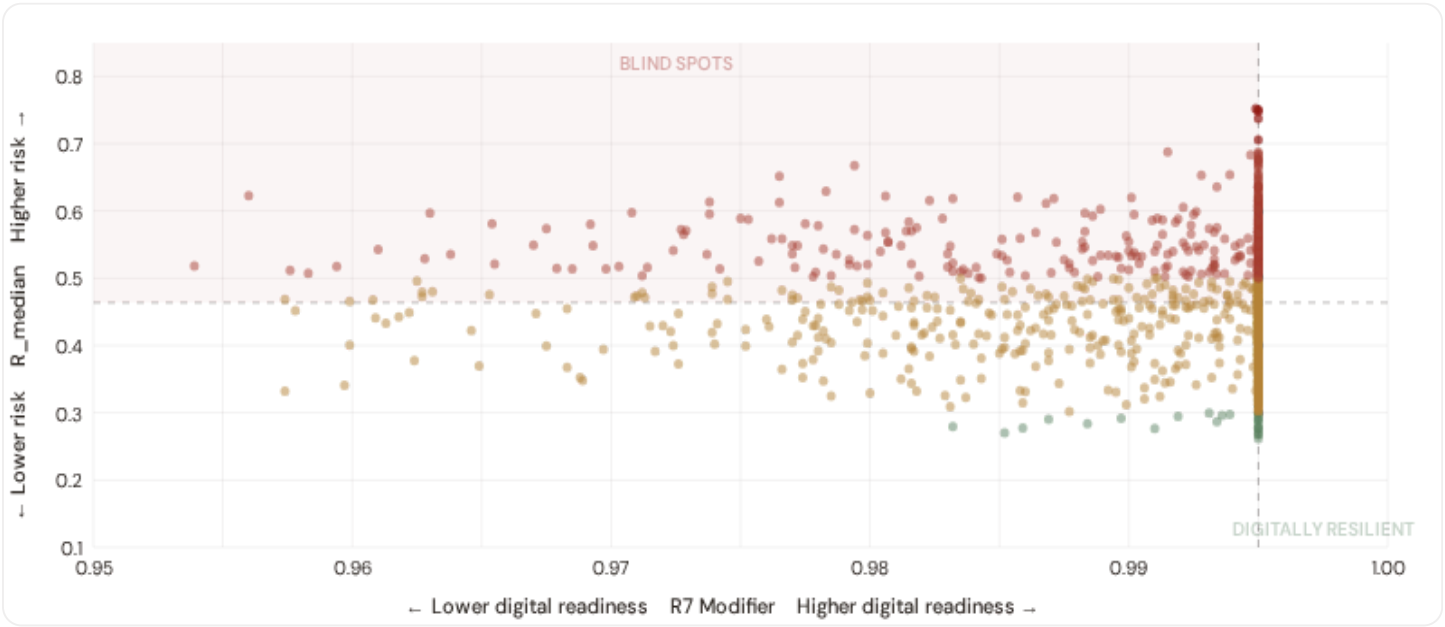
HIGH-CONSEQUENCE SUBS

204

13.1% · R3 ≥ 1.04

B.1 Cyber-Physical Risk Matrix

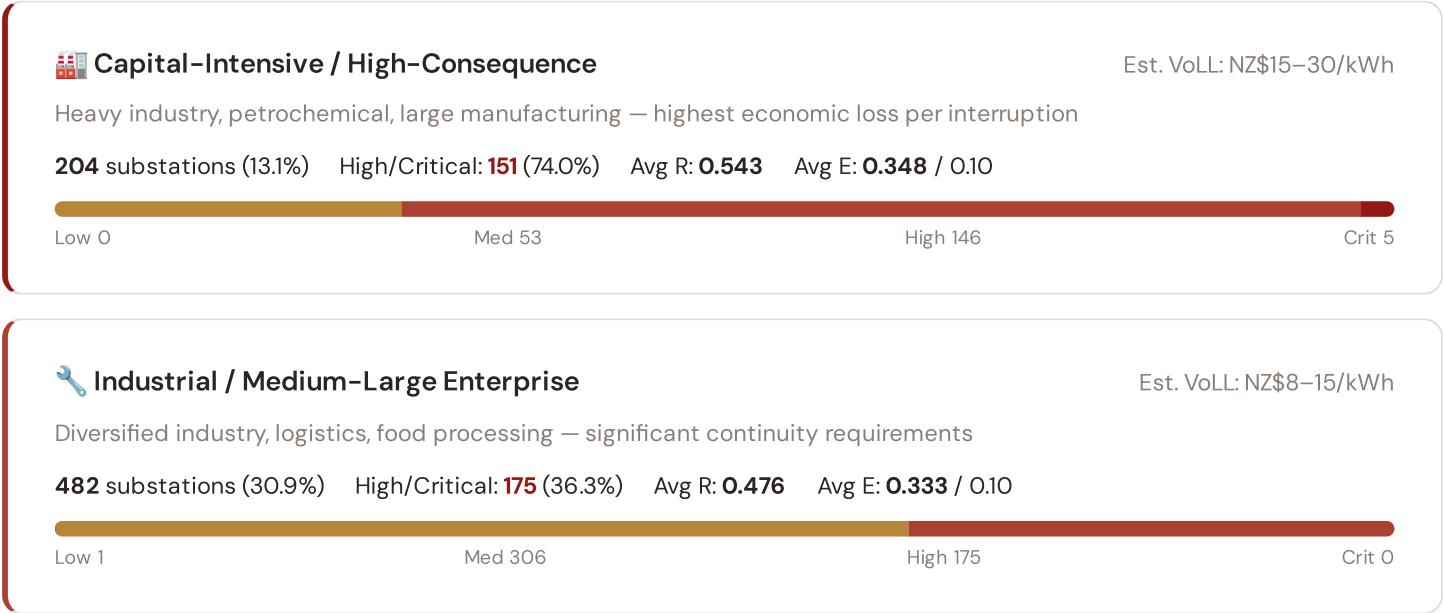
Each substation plotted by R7 digital readiness (x-axis) vs overall R_{median} (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.



The cyber-physical matrix identifies **184 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness (R7 < 1.0000). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **Wellington (88)**, **Canterbury (55)**, **Auckland (17)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 220 (14.1%) are in the LOW tier — indicating strong digital infrastructure coverage but concentrated in the northern urban grid.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / SME-Dense

Est. VoLL: NZ\$3–8/kWh

Service economy, retail, tourism — moderate individual impact but high aggregate exposure

495 substations (31.8%) High/Critical: **138** (27.9%) Avg R: **0.455** Avg E: **0.321** / 0.10



Light / Agricultural / Rural

Est. VoLL: NZ\$1–3/kWh

Sparse economic activity, agricultural land — lower VoLL but higher social vulnerability

377 substations (24.2%) High/Critical: **52** (13.8%) Avg R: **0.423** Avg E: **0.328** / 0.10



Value of Lost Load (VoLL) context: The Electricity Authority's regulatory framework places New Zealand's average VoLL at approximately NZ\$12.5/kWh for industrial customers and NZ\$3.8/kWh for residential — reflecting mid-high developed-country GDP per capita. The SSI's R3 consequence multiplier identifies which substations serve the most economically exposed territories. **204 metropolitan substations** ($R3 \geq 1.04$) concentrate in the Auckland–Wellington corridor, of which **151** also carry High or Critical risk — representing the highest VoLL-weighted exposure in the fleet.

The metropolitan tier concentrates in **Wellington (174)**, **Northland (23)**, **West Coast (7)** — where financial services, technology, port logistics, and the Fonterra/dairy industry automotive supply chain depend on uninterrupted power. New Zealand's economic fabric is heavily SME-driven (~99.9% of firms), meaning outage costs propagate through dense networks of small businesses rather than a few large plants. The fleet-wide average energy poverty rate is **6.2%** — significantly above the developed-country average (~5%) — with sharp regional variation: rural regions combine the highest energy poverty with aging grid infrastructure, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is a distinctive feature of the SSI — linking VoLL exposure to substation-level risk in a way that conventional frameworks do not.

B.3 Region Digital Readiness Ranking

Regions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_{median} to identify regions combining digital exposure with high grid stress.



WORST DIGITAL READINESS — TOP 15 PROVINCES

Longer bar = higher R7 = better digital infrastructure.

1	Gisborne	0.9944
2	West Coast ▲ high R	0.9947
3	Taranaki	0.9957
4	Wellington ▲ high R	0.9958
5	Bay of Plenty	0.9960
6	Canterbury ▲ high R	0.9960
7	Marlborough ▲ high R	0.9962
8	Auckland	0.9965
9	Hawke's Bay ▲ high R	0.9967
10	Manawatu-Whanganui	0.9973
11	Waikato	0.9973
12	Otago	0.9978
13	Southland	0.9979
14	Tasman	0.9981
15	Northland	0.9985

Region-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **Gisborne** has the lowest average R7 (0.9944), while **Northland** leads at 0.9985 — a spread of 0.0042 across the R7 range. **5 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for New Zealand 2030 / New Zealand Reliance / PPE digitalisation investment. The R7 modifier is intentionally narrow ([0.99, 1.05]) to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (April 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9965, median = 1.0000; 0 substations classified HIGH cyber-exposure; 184 blind-spot substations (High/Critical risk + below-median R7); 1 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline, flagging regions where DESI improvements (broadband rollout, smart meter deployment, OT upgrades) translate into measurable R7 shifts. The next material R7 update is expected when Eurostat publishes the 2025 DESI sub-indices (anticipated Q3 2026).

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **28 verified public data sources** feeding 95 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.

TOTAL SOURCES

28

WEEKLY / MONTHLY

3

QUARTERLY / ANNUAL

16

95 variables

High-frequency feeds

Regular refresh cycle

STATIC / DERIVED

2

Standards + scenarios

Data Source Registry — April 2026

OSM	OpenStreetMap — Power Infrastructure	WEEKLY	Node/edge	3 vars
EA	Electricity Authority (EA)	MONTHLY	National	5 vars
EECA	EECA (Energy Efficiency & Conservation Authority)	QUARTERLY	Regional	4 vars
NZTA	NZTA (NZ Transport Agency)	QUARTERLY	Regional	1 var
MBIE	MBIE (Ministry of Business, Innovation & Employment)	ANNUAL	Regional	4 vars
ME	Ministry for the Environment	ANNUAL	Regional	3 vars
CC	Commerce Commission	ANNUAL	TA/Regional	3 vars
StatsNZ	Stats NZ (Tatauranga Aotearoa)	ANNUAL	TA	8 vars
Vector	Vector Ltd — Largest EDB	ANNUAL	Network	2 vars
Orion	Orion NZ — Canterbury/Christchurch Distribution	ANNUAL	Network	2 vars
Powerco	Powerco — Waikato/Taranaki/Manawatu Distribution	ANNUAL	Network	2 vars
Meridian	Meridian Energy — Largest Generator	ANNUAL	Regional	2 vars
Genesis	Genesis Energy — Thermal/Hydro/Wind	ANNUAL	Regional	2 vars
Contact	Contact Energy — Geothermal/Gas Generation	ANNUAL	Regional	2 vars
Mercury	Mercury NZ — Waikato Hydro/Geothermal	ANNUAL	Regional	2 vars
FE-NZ	Fire and Emergency NZ	ANNUAL	Regional	2 vars
EQC	Earthquake Commission (EQC)	ANNUAL	Regional	2 vars
LGNZ	LGNZ (Local Government NZ)	ANNUAL	TA	2 vars
BRANZ	BRANZ (Building Research Association)	ANNUAL	National	2 vars
EDB-Disc	EDB Information Disclosures	ANNUAL	EDB	3 vars
LINZ	LINZ (Land Information New Zealand)	STATIC	National	3 vars
Copernicus	ERA5 / Copernicus Climate Data Store	STATIC	0.25° (~27 km)	4 vars
GeoTech	NZ Geotechnical Database	STATIC	Site-level	2 vars
TPNZ	Transpower NZ — TSO	REAL-TIME	GXP/Regional	8 vars
NIWA	NIWA (National Institute of Water & Atmospheric Research)	HOURLY	~25 km	4 vars
GeoNet	GNS Science / GeoNet	REAL-TIME	Event-level	3 vars
SO	Transpower System Operator (SO)	REAL-TIME	Real-time	3 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Weekly (2 sources)

OSM diffs + weather reanalysis

OSM Overpass · MetService/NIWA

Monthly (1 sources)

Grid load + procurement data

EA Market Data

Quarterly (2 sources)

DER registry + business data + macro indicators

EECA Surveys · NZTA Infrastructure

Annual (14 sources)

Core grid quality + demographics + environment

StatsNZ · MBIE · CC · EA · Vector · Orion · Powerco · Meridian · Genesis · Contact · Mercury · Fire & Emergency · EQC · EDB-Disc

Static (4 sources)

Standards + scenarios + seismic hazard

Copernicus CMIP6 · GeoTech Database · LINZ Topography · ISO 9223

C.2 Fleet Score Snapshot

FLEET SIZE

1,558

substations scored

MEDIAN R

0.464

P5=0.333 · P95=0.602

HIGH + CRITICAL

516

33.1% of fleet

HIGH CONFIDENCE

100%

1558 subs · CI/R < 0.50

BAND DISTRIBUTION

Low 26 (1.7%)

Medium 1016 (65.2%)

High 511 (32.8%)

Critical 5 (0.3%)

No primary data sources have been updated since the v4.0.2 launch baseline. All **1,558 substation scores remain unchanged** this edition. The fleet median R of 0.464 (mean 0.465) reflects a distribution skewed toward the lower bands, with 1.7% in Low and 65.2% in Medium. The first material score changes are expected when Electricity Authority publishes the annual TIQE indicators (affecting C1–C4 and R6a) and when MBIE refreshes the Electricity Registry (affecting T1). High-frequency sources (OSM, Open-Meteo) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

17 changes tracked across P0, P1, and P2 releases

F1	NEW	New T component — Energy Transition Exposure (T1)	\$2, \$4
F2	NEW	New R6a — Restoration Speed Modifier (CAIDI-based)	\$5
F3	ENHANCED	R3 enhanced — Social Vulnerability Scoring (V_socio)	\$5
F4	ENHANCED	R4 enhanced — Graph-theoretic betweenness + bridge detection	\$5
F5	ENHANCED	R2 enhanced — Climate Trajectory (CMIP6 SSP2-4.5)	\$5
F6	ENHANCED	R7 Digital Readiness — Regional-level capability model	\$5
L1	NEW	New I5, I7–I9 metrics — thermal, load, corrosion, hydrogeo	\$2, \$8
L2	ENHANCED	E2 Economic Enrichment — Business structure + innovation capacity	\$6
L3	ENHANCED	V_socio Fiscal Enrichment — LGNZ + CC + energy price data	\$5
L4	ENHANCED	R3 Demographic Shift Amplifier — StatsNZ net migration	\$5
L5	DATA	95/95 variables operational (100%). 28 data sources total.	\$8, \$12
G1	DATA	TPNZ upgraded to real-time API — grid status and dispatch data	\$12
G2	DATA	OSM upgraded to Overpass API — ~11,500 real substations, 12,400 lines	\$12
G3	DATA	GeoNet seismic monitoring — real-time PGA + historical catalog	\$12
G4	NEW	R6b Seismic modifier — variable α by region (Alpine Fault 0.70, Wellington 0.60, Auckland 0.15)	\$5
G5	NEW	Network & Topology layer (H): 7 variables — TPNZ grid plan, ring analysis	\$12
G6	NEW	Output Scores layer (I): 7 variables — risk_score, ETTTC, stationary probs	\$12

SECTION D — MONTHLY DEEP-DIVE

The Waikato River Valley Corridor: Where Seismic Risk Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation: **Ed. 001** Waikato River Valley corridor · **002** DER saturation hotspots (Waikato solar belt) · **003** Climate vulnerability under CMIP6 · **004** Metropolitan–Rural economic impact disparity · **005** Seismic risk and West Coast isolation · **006** Infrastructure age vs smart metering rollout · **007** Stress–Resilience quadrant trends · **008** T-component forward projections · **009** Region Grid Health Cards · **010** Academic validation and peer feedback · **011** OECD replication feasibility · **012** Year in review.

D.1 Why the Waikato River Valley, Why Now

WAIKATO RIVER VALLEY
SUBSTATIONS

91

HIGH BAND

18

19.8% of corridor

CORRIDOR MEDIAN R

0.465

Fleet median: 0.464

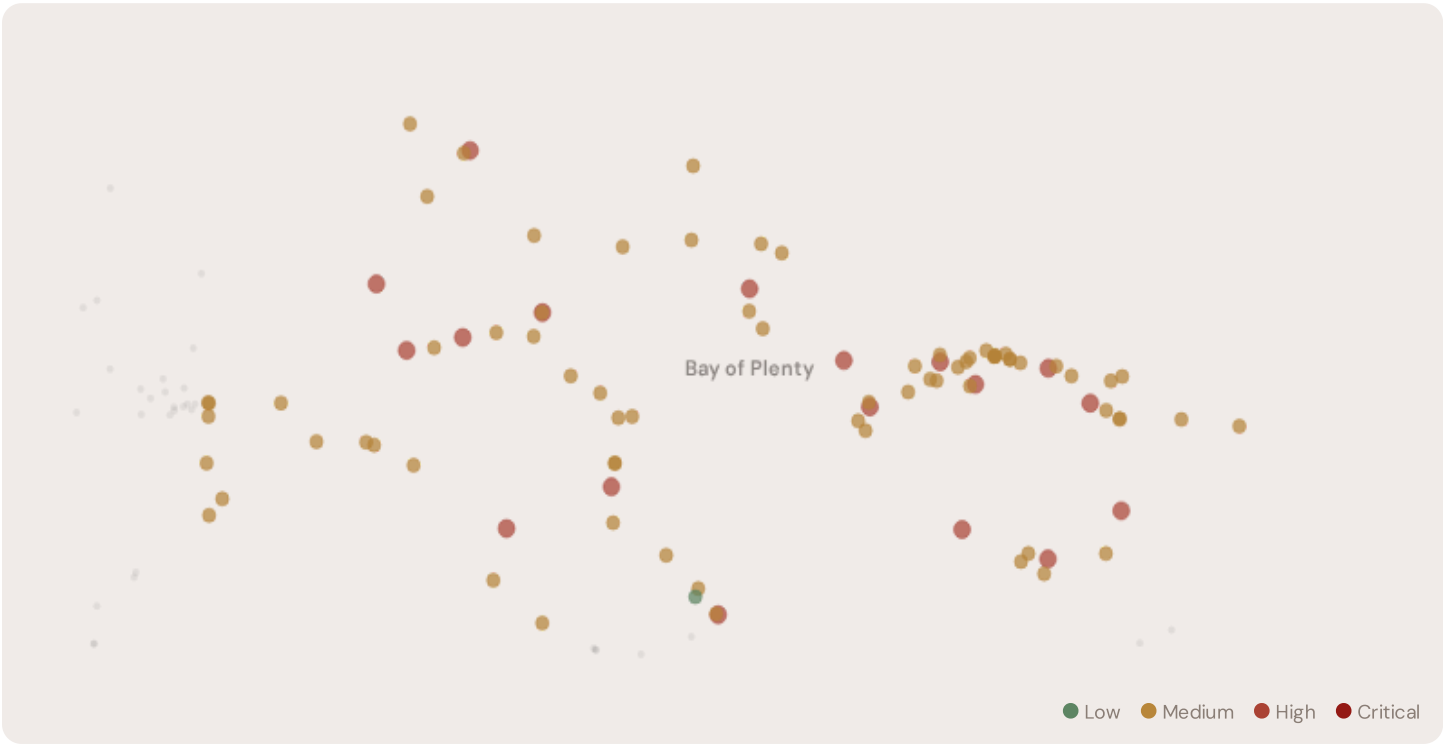
WORST REGION

Bay of Plenty

Avg R: 0.461 · 91 substations

The Waikato River Valley hosts **91 substations** across six regions, making it one of the most energy-critical corridors in New Zealand. However, its risk profile is disproportionately concentrated in the upper bands: **18 substations (19.8%)** are classified High, with only **1** in the Low band. 53% of Waikato River Valley substations score above the national fleet median of 0.464. The corridor median R of 0.465 reflects the tension between seismic exposure, rapidly growing renewable variability, wildfire risk, and aging grid infrastructure spanning New Zealand's interior. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under New Zealand 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Otumoetai zone substation	Bay of Plenty	0.570	High	0.463	0.490	0.509	0.187	0.501	0.541	
Arawa Substation	Bay of Plenty	0.565	High	0.597	0.411	0.367	0.497	0.687	0.257	S
Te Maunga zone substaion	Bay of Plenty	0.564	High	0.558	0.289	0.506	0.361	0.697	0.395	S
Kauri Point zone substation	Bay of Plenty	0.559	High	0.460	0.348	0.595	0.469	0.542	0.314	

SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
Bay of Plenty_1424946503	Bay of Plenty	0.558	High	0.619	0.315	0.508	0.466	0.441	0.514	C
Ōmokoroa Zone Substation	Bay of Plenty	0.555	High	0.353	0.254	0.834	0.219	0.557	0.310	I
Okere Substation	Bay of Plenty	0.545	High	0.375	0.425	0.447	0.451	0.451	0.611	T
Kopu Substation	Bay of Plenty	0.541	High	0.400	0.455	0.649	0.441	0.425	0.365	I
Mikkelsen zone substation	Bay of Plenty	0.535	High	0.609	0.394	0.489	0.360	0.485	0.582	C
Tahuna Substation	Bay of Plenty	0.528	High	0.414	0.269	0.415	0.399	0.852	0.341	S

... 81 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — WAIKATO RIVER VALLEY VS FLEET			MODIFIER AVERAGES — WAIKATO RIVER VALLEY VS FLEET		
C — Continuity	0.453 vs 0.473 (−4%)		R3 Consequence	0.995	fleet 0.990 →
I — Infrastructure	0.415 vs 0.424 (−2%)		R4 Graph Crit.	1.000	fleet 0.997 →
S — Saturation	0.502 vs 0.449 (+12%)		R6a Restoration	0.993	fleet 0.992 →
V — Voltage	0.323 vs 0.342 (−6%)		R6b Seismic	1.103	fleet 1.119 ↑
E — Economic	0.333 vs 0.330 (+1%)		R7 Digital Read.	0.996	fleet 0.996 →
T — Transition	0.389 vs 0.397 (−2%)				

The dominant risk driver across the Waikato River Valley corridor is **T (Transition)**, averaging 0.389 against a fleet average of 0.397 — occupying 778% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.333 vs fleet 0.330 (333% of ceiling). Among modifiers, the R3 consequence multiplier averages 0.995 (fleet: 0.990), reflecting the socio-economic fragility of The Waikato River Valley's mixed economy with significant hydroelectric capacity and growing solar penetration. The R6b seismic modifier averages 1.103, capturing the region's moderate but non-negligible seismic exposure — a dimension invisible to every competing framework.

D.4 Region Breakdown

PROVINCE RISK PROFILE — SORTED BY MEAN R	
Longer bar = lower R = better resilience.	
Bay of Plenty	91 subs · avg R = 0.461 · H:18 M:72

The province-level breakdown reveals significant intra-regional variation. **Bay of Plenty** leads with a mean R of 0.461 across 91 substations, while **Bay of Plenty** scores 0.461 — a spread of 0.000 within a single region. This disparity underscores why regional-level metrics (as used by CEER and JRC) mask the true distribution of grid stress. The SSI's substation-level granularity reveals that the Waikato River Valley is not uniformly stressed but contains distinct corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

O Waikato River Valley substations score $R \geq 0.650$, placing them in the upper quartile of national risk. For PPE (Programmation Pluriannuelle de l'Énergie) investment prioritisation, this analysis identifies the Bay of Plenty corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve — characterised by service-economy dependence, tourism exposure, and above-average energy poverty — are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

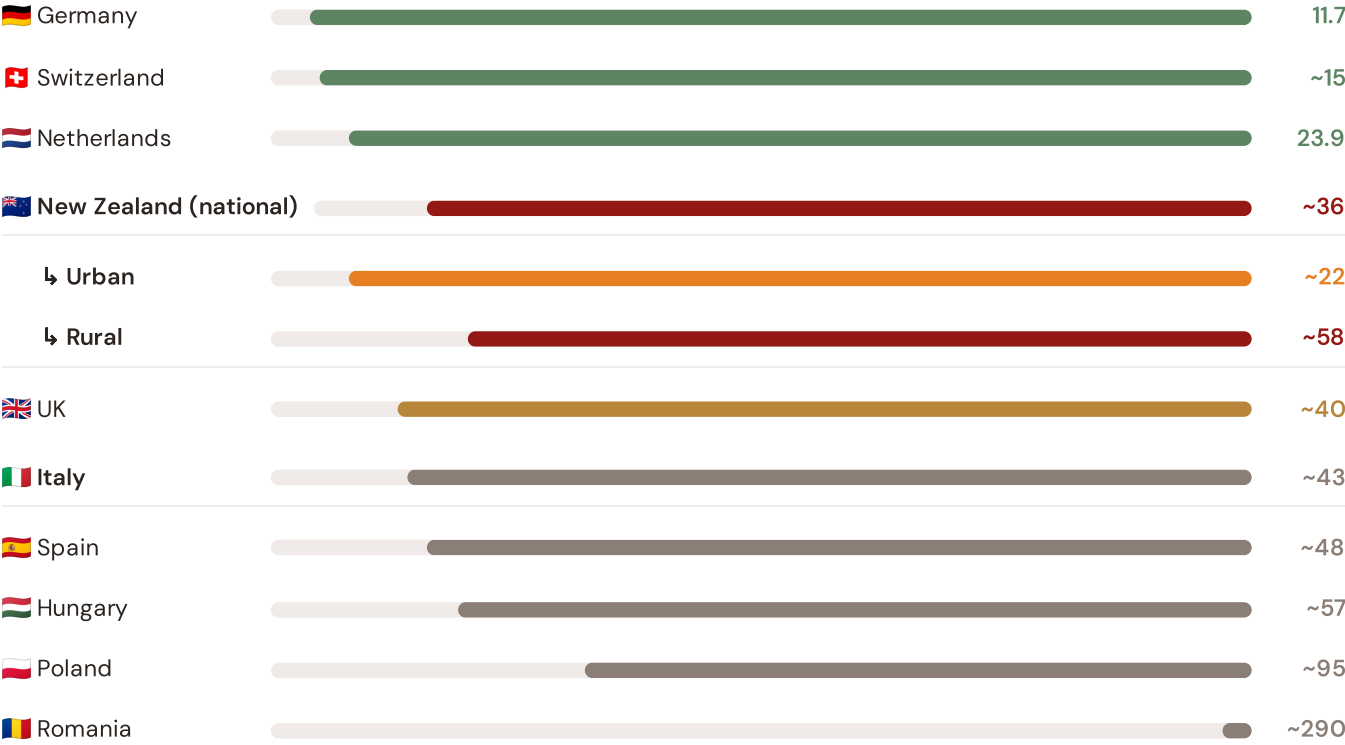
SECTION E — RECURRING MODULE

OECD Context Card

How this works: Each edition includes one OECD Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Waikato River Valley corridor analysis hinges on New Zealand's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



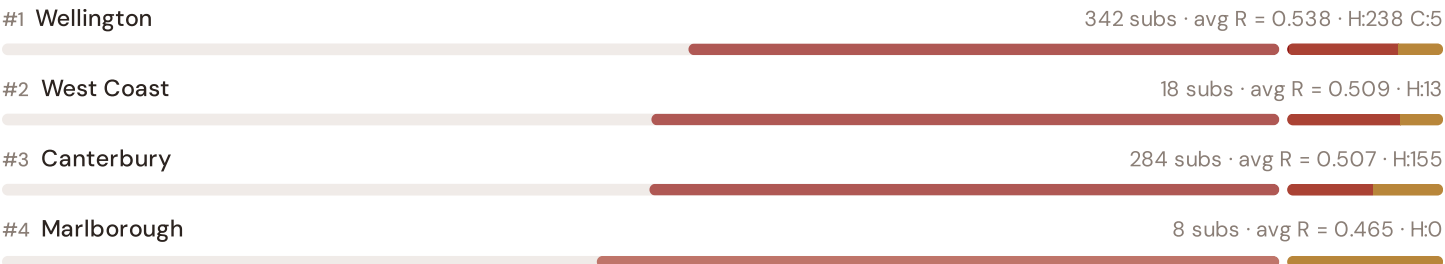
Source: CEER 7th Benchmarking Report (2022); Electricity Authority TIQE (2023); Bundesnetzagentur (2024). New Zealand sub-categories from Electricity Authority territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats.
· Updated annually with CEER BR release.

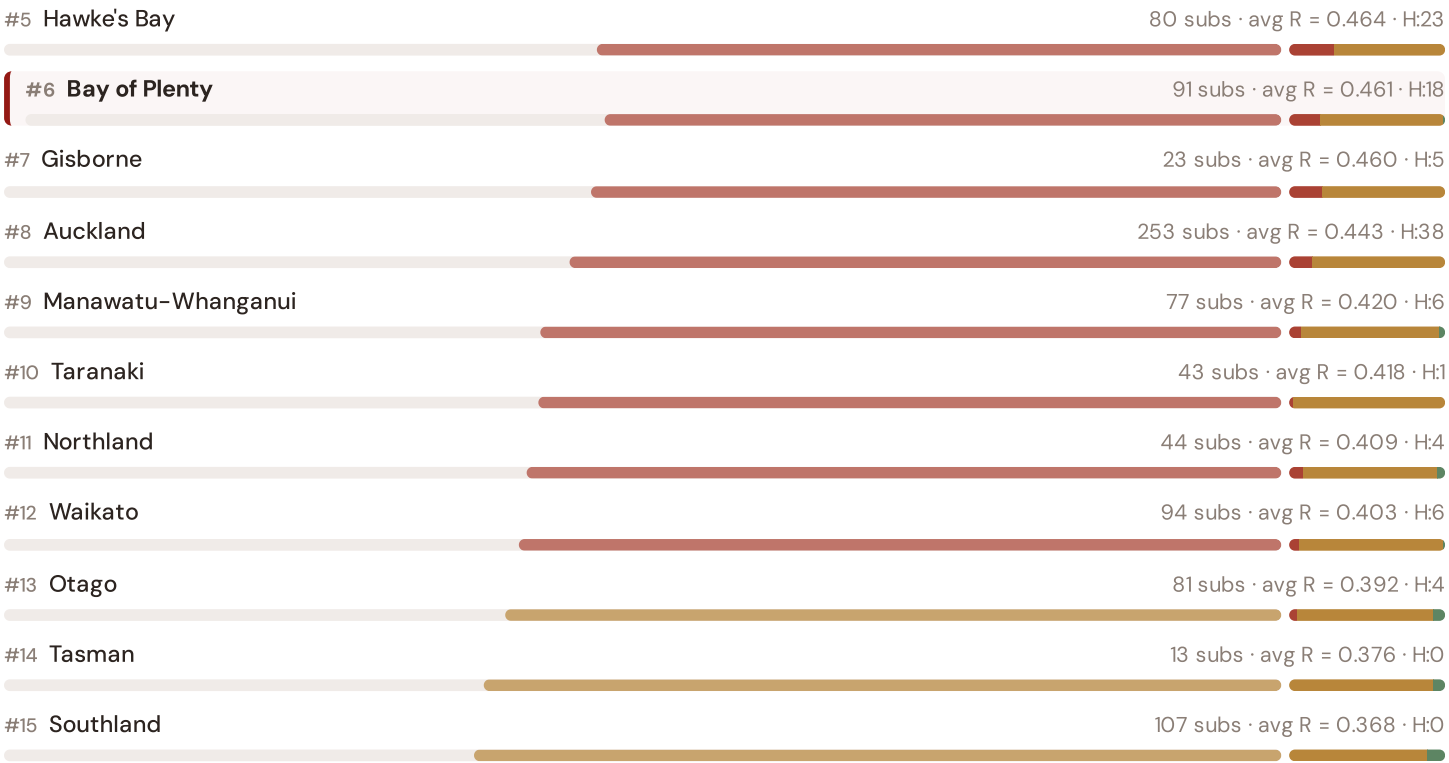
This month's key insight: The Waikato River Valley corridor deep-dive shows **91 substations** (of 91) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with New Zealand's rural territory classification. The Waikato River Valley's average C of 0.453 is **1.0× the fleet average** of 0.473. In Pacific context, these substations individually perform closer to regional SAIDI levels (~90–120 min/customer/year) than to New Zealand's own national average (~180 min). The SSI reveals what aggregate Electricity Authority statistics conceal: that New Zealand's reliability position is not a national condition but a statistical average of Auckland-quality urban performance and remote-rural-level performance — and the Waikato River Valley corridor concentrates the worst of the latter.

Regional Risk Ranking — All 7 New Zealand Islands

Where does the Waikato River Valley sit within the national landscape? Regions sorted by mean R_{median}, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



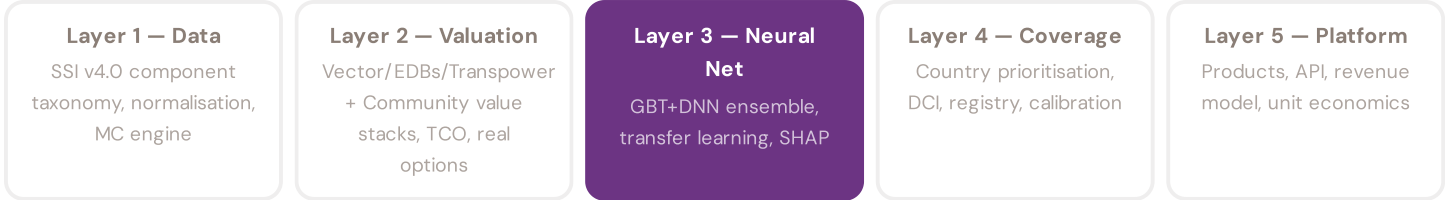


The Waikato River Valley ranks **#6 of 7 islands** by mean R_{median} , placing it among the upper tier of national risk. The three most stressed regions are Wellington, West Coast, Canterbury, while the three most resilient are Southland, Tasman, Otago. 3 of 7 islands score above the fleet average of 0.465. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress across New Zealand. It is precisely this substation-level granularity that positions the SSI as a tool for PPE / New Zealand 2030 / New Zealand Reliance investment prioritisation: not treating entire islands as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Vector/EDBs/Transpower and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · Inaugural edition · April 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. A BESS developer considering a NZ\$15M commitment needs to understand not just that a substation scores highly, but which specific factors drive that score — and whether those factors are likely to persist. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the BESS value depends primarily on the substation's poor reliability history — and can verify this against Electricity Authority (EA)'s published TIQE data. Transparency is not optional; it is what converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall visualisation: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The New Zealand training set comprises 1,558 substations with complete 95-feature vectors. Average CI_width of 0.131 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 26 Low, 1016 Medium, 511 High, 5 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 002

The Canterbury Seismic Coastline

Deep-dive into Canterbury's grid risk profile — substation map, component analysis, province breakdown. OECD Context Card: Alpine Fault / Hikurangi subduction exposure combined with extreme seasonal tourism demand peaks.

NEXT DATA REFRESH

Q3 2026 — 2 Quarterly Sources

EECA Surveys, NZTA Infrastructure are due for refresh in Q3. Impact: MBIE (Electricity Registry) data refresh schedule (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **Stats NZ (Tatauranga Aotearoa)** (8 variables → Census, GDP, population, employment, household income).

FLEET SPOTLIGHT

0 Substations Approaching Critical Degradation

The SSI's 4-state Markov degradation model identifies **0 substations** with a stationary critical probability above 70% — meaning their long-term equilibrium state is dominated by degraded or critical condition. Without intervention, these substations will trend toward failure. Average estimated time-to-critical: ? years.

Edition 002 will be published on the second Thursday of June 2026. Deep-dive region: **Canterbury**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a New Zealand utility, Electricity Authority, Transpower, Vector/EDBs, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Edition 001 (Inaugural)

Published 9 April 2026 · SSI Index v4.0.2 · 1,558 substations · ~1,686 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5